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Course and Section: BSIT DA 2-1

W1-Quiz1 Activity: Introduction to Algorithms

A. Discuss Briefly

1. What is an algorithm, and how is it related to the tasks you perform in your daily life?

An **algorithm** is a step-by-step set of instructions used to solve a problem or complete a task. In daily life, we use algorithms even without realizing it. For example, when cooking, we follow steps such as preparing the ingredients, mixing them, cooking the food, and serving it. Following these steps in the correct order helps us complete the task properly.

2. Why is it important for an IT student to learn algorithms?

Learning algorithms is important for an IT student because it helps develop **logical and problem-solving skills**. Algorithms teach students how to break a problem into smaller and easier steps. They are also important in programming because programmers use algorithms to create programs that work correctly and efficiently.

B. Convert each JavaScript code into a step-by-step algorithm using simple English or pseudocode.

***Guide**: Describe what the program does from the beginning until the loop stops. You may use numbered steps.*

1.
```
for (let i = 1; i < 10; i *= 2) {
    console.log(i);
}
```

1. Start with i = 1.
2. Check if i is less than 10.
3. If true, display the value of i.
4. Multiply i by 2.
5. Check again if i is less than 10.
6. Repeat the process until i is no longer less than 10.
7. Stop the loop

Output: 1, 2, 4, 8

```
for (let i = 0; i < 10; i++) {
2.    console.log(i);
}
```

Ans:

1. Start with i = 0.
2. Check if i is less than 10.
3. If true, display the value of i.
4. Add 1 to i.
5. Check again if i is less than 10.
6. Repeat the process until i reaches 10.
7. Stop the loop.

Output: 1, 2, 3, 4, 5, 6, 7, 8, 9

C. Observation and Analysis

1. What did you observe about the sequence of numbers produced by the first code?

The first code produces numbers that double each time. The sequence is 1, 2, 4, 8. Instead of increasing by a fixed amount, the number is multiplied by 2 after every loop.

2. What did you observe about the sequence of numbers produced by the second code?

The second code produces numbers in increasing order, starting from 0 up to 9. The number increases by exactly 1 each time.

3. What is the main difference between the way the two loops progress?

The first loop uses i *= 2, so the valu of i is multiplied by 2 after every repetition. The second loop uses i++, so the value of i is increased by 1 afer every repetition.

4. How does the way an algorithm performs its steps affect its output?

The way an algorithm performs its steps directly affects its output. If the steps or the way the values change are different, the result will also be different. In these examples, the first algorithm doubles the value each time, while the second increases it by one. Therefore, they produce different sequences even though both use a for loop.